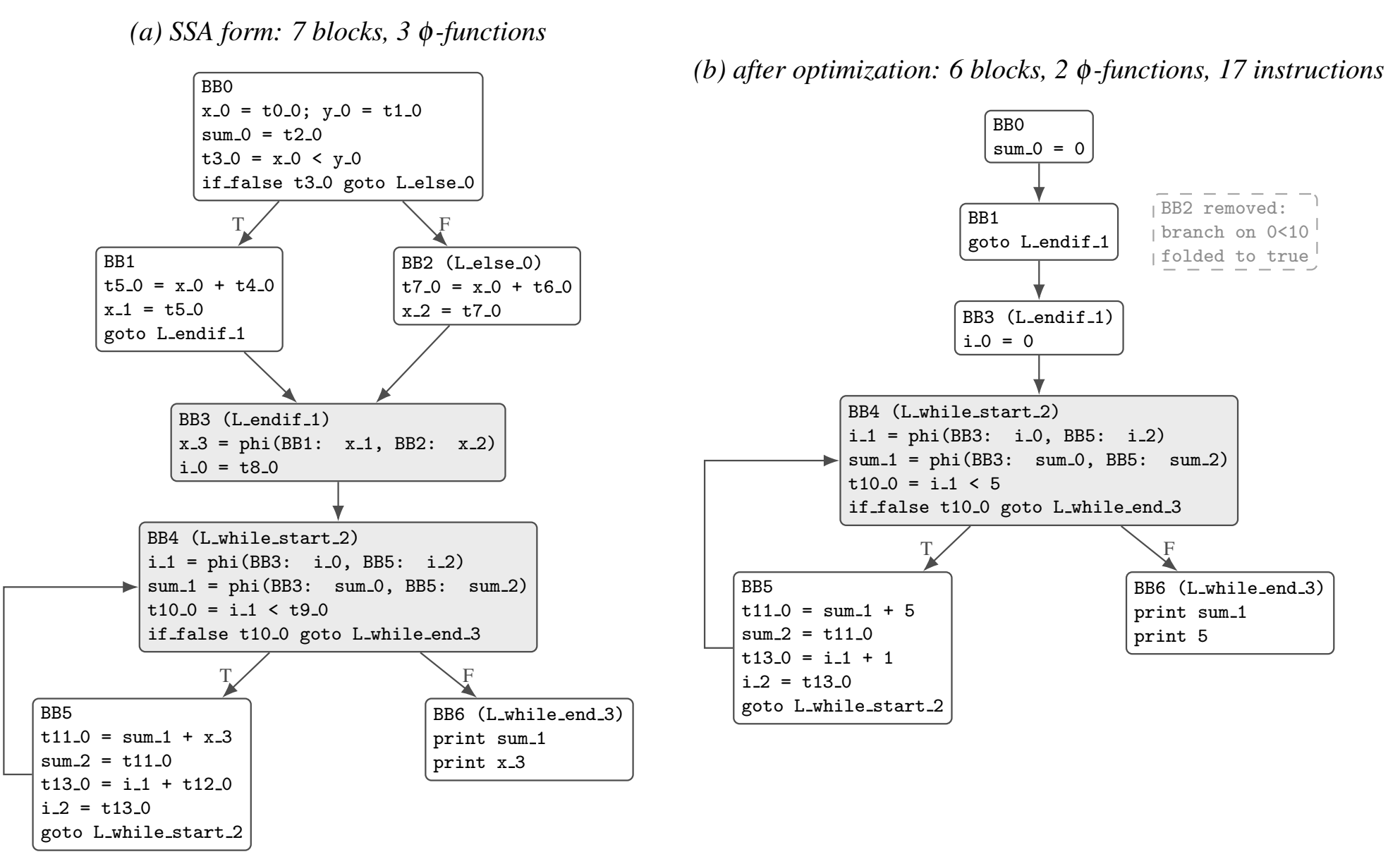
(a) SSA form: 7 blocks, 3 ϕ-functions
BB0
x_0 = t0_0; y_0 = t1_0
sum_0 = t2_0
t3_0 = x_0 < y_0
if_false t3_0 goto L_else_0
T
F
BB1
t5_0 = x_0 + t4_0
x_1 = t5_0
goto L_endif_1
BB2 (L_else_0)
t7_0 = x_0 + t6_0
x_2 = t7_0
BB3 (L_endif_1)
x_3 = phi(BB1: x_1, BB2: x_2)
i_0 = t8_0
BB4 (L_while_start_2)
i_1 = phi(BB3: i_0, BB5: i_2)
sum_1 = phi(BB3: sum_0, BB5: sum_2)
t10_0 = i_1 < t9_0
if_false t10_0 goto L_while_end_3
T
F
BB5
t11_0 = sum_1 + x_3
sum_2 = t11_0
t13_0 = i_1 + t12_0
i_2 = t13_0
goto L_while_start_2
BB6 (L_while_end_3)
print sum_1
print x_3
(b) after optimization: 6 blocks, 2 ϕ-functions, 17 instructions
BB0
sum_0 = 0
BB1
goto L_endif_1
BB2 removed:
branch on 0<10
folded to true
BB3 (L_endif_1)
i_0 = 0
BB4 (L_while_start_2)
i_1 = phi(BB3: i_0, BB5: i_2)
sum_1 = phi(BB3: sum_0, BB5: sum_2)
t10_0 = i_1 < 5
if_false t10_0 goto L_while_end_3
T
F
BB5
t11_0 = sum_1 + 5
sum_2 = t11_0
t13_0 = i_1 + 1
i_2 = t13_0
goto L_while_start_2
BB6 (L_while_end_3)
print sum_1
print 5